Assignment -12

Given Parameters

* States: S_0, S_1 + Goal
* Actions: Up, Down, left + Right
* Discount factor: $\gamma = 0.9$
* Normal movement reward: $R = -1$
* Goal = +100
* Terminal state value: $V(G) = 0$
* Convergence threshold $\theta = 0.001$

Transition Table:

S_0	-1
S_1	-1
S_0	-1
S_0	-1
S_0	-1
S_1	-1
S_1	-1
G	+100

Value Iteration formula:

Bellman Optimality

$$V_{k+1}(s) = \max_a [R(s, a, s') + \gamma V_k(s')]$$

where

* $V_k(s)$ = Current state value
* $V_{k+1}(s)$ = Updated state value
* $R(s, a, s')$ = Immediate Reward
* γ = discount factor
* s' = Next State

Terminal State: $V(G) = 0$

Initially ⇒ $V_0(s_0) = 0$ $V_0(s_1) = 0$, $V_0(G) = 0$

Iteration -1:

$V_0(s_0) = 0$, $V_0(s_1) = 0$

for state s_0

⇒ Up :- $Q(s_0, Up) = -1 + 0.9(0) = -1$

⇒ Down :- $Q(s_0, Down) = -1 + 0.9(0) = -1$

⇒ left :- $Q(s_0, left) = -1 + 0.9(0) = -1$

⇒ Right :- $Q(s_0, Right) = -1 + 0.9(0) = -1$

Therefore.

$V_1(s_0) = \max(-1, -1, -1, -1) \Rightarrow \boxed{V_1(s_0) = -1}$

for state s_1 :-

⇒ Up :- $Q(s_1, Up) = -1 + 0.9(0) = -1$

⇒ down :- $Q(s_1, down) = -1 + 0.9(0) = -1$

⇒ left :- $Q(s_1, left) = -1 + 0.9(0) = -1$

⇒ Right :- $Q(s_1, Right) = 100 + 0.9(0) = 100$

Therefore $V_1(s_1) = \max(-1, -1, -1, 100) \Rightarrow \boxed{V_1(s_1) = 100}$

The final optimal state

$V^*(s_0) = 89$

$V^*(s_1) = 100$

$V^*(G) = 0$

The optimal path ⇒ $\boxed{s_0 \rightarrow s_1 \rightarrow G}$